

OPOLE, Poland

WMO: 125300

Lat: 50.6269N

Lon: 17.9689E

Elev: 163

StdP: 99.38

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-10.9	-16.9	0.9	-13.6	-14.1	1.1	-10.2	9.2	3.0	8.2	2.7	1.5	90	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.2	20.4	29.1	19.5	27.3	18.7	21.4	28.4	20.6	27.0	19.7	25.4	2.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	23.5	18.4	13.5	22.7	17.7	12.9	22.0	62.8	28.7	59.8	27.1	56.9	25.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.4	5.6	-16.8	34.2	4.0	1.6	-19.6	35.3	-22.0	36.3	-24.2	37.2	-27.1	38.4		
(5)			-17.1	23.3	3.9	1.3	-19.9	24.2	-22.2	25.0	-24.4	25.7	-27.3	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-0.9	0.7	4.1	9.6	14.1	17.6	19.7	19.4	14.6	9.8	5.1	0.7		
	DBStd	8.31	5.27	5.12	4.28	4.25	3.73	3.44	3.15	3.13	3.45	4.16	4.24	4.81		
	HDD10.0	1347	337	261	186	57	7	0	0	0	3	56	153	289		
	HDD18.3	3357	594	494	440	263	138	54	21	25	121	264	396	547		
	CDD10.0	1201	0	1	4	45	135	228	301	290	141	50	7	1		
	CDD18.3	168	0	0	0	1	8	32	63	57	7	1	0	0		
	CDH23.3	1756	0	0	0	21	112	344	644	549	84	1	0	0		
	CDH26.7	527	0	0	0	2	16	98	222	172	17	0	0	0		
(14) Wind	WSAvg	2.6	2.9	2.9	3.0	2.8	2.5	2.3	2.2	2.0	2.2	2.4	2.7	2.9		
(15) Precipitation	PrecAvg	580	32	29	37	38	60	73	91	57	56	39	39	32		
	PrecMax	722	57	70	83	64	115	168	277	110	104	89	73	98		
	PrecMin	377	11	3	13	5	23	17	7	12	14	4	0	12		
	PrecStd	80	13	16	18	15	24	35	62	23	27	21	18	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	13.5	18.4	26.0	28.6	32.3	33.9	33.9	29.3	23.1	16.7	12.3		
		MCWB	8.3	9.0	11.1	15.9	17.8	21.4	20.4	21.2	18.4	16.2	12.3	9.3		
	2%	DB	8.9	11.0	15.5	22.5	26.4	29.5	31.5	30.6	25.7	20.5	14.9	10.1		
		MCWB	6.9	7.7	9.7	13.7	17.6	20.1	20.1	20.3	17.3	15.0	11.4	7.8		
	5%	DB	7.2	9.3	13.1	19.8	24.2	27.1	29.1	28.4	23.5	18.4	13.1	8.1		
		MCWB	5.3	6.7	8.6	12.3	16.4	18.7	19.5	19.4	16.5	14.2	10.3	6.4		
	10%	DB	5.8	7.6	10.9	17.6	21.9	25.0	27.1	26.4	21.3	16.2	11.4	6.5		
		MCWB	4.2	5.3	7.4	11.3	15.2	17.8	18.9	18.6	15.8	13.0	9.2	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	9.6	11.8	16.5	19.7	22.4	22.5	22.3	19.6	16.9	13.0	9.6		
		MCDB	10.5	12.2	17.2	23.8	25.9	29.8	29.9	29.7	25.9	21.3	16.0	11.9		
	2%	WB	6.9	8.2	10.3	14.7	18.2	20.9	21.4	21.1	18.4	15.6	11.7	8.0		
		MCDB	8.6	10.7	14.4	21.2	24.3	27.8	28.2	28.3	23.8	19.5	14.4	9.9		
	5%	WB	5.5	6.8	9.0	13.1	17.1	19.7	20.6	20.2	17.4	14.5	10.5	6.5		
		MCDB	7.0	9.0	12.4	18.3	22.8	25.4	26.7	26.7	22.0	17.7	12.7	8.0		
	10%	WB	4.4	5.4	7.6	11.8	16.0	18.6	19.7	19.4	16.4	13.3	9.3	5.2		
		MCDB	5.7	7.4	10.6	16.6	20.8	23.6	25.3	24.7	20.3	15.8	11.2	6.4		
(35) Mean Daily Temperature Range	MDBR		5.2	6.3	8.2	10.5	10.7	10.5	10.8	11.2	9.7	8.0	5.8	5.0		
	5% DB	MCDBR	6.0	8.6	12.4	14.9	14.8	14.6	15.1	14.8	13.8	11.3	8.5	6.1		
		MCWBR	4.6	5.9	7.5	7.5	7.0	6.3	5.6	5.8	6.5	6.6	5.6	4.7		
	5% WB	MCDBR	5.9	8.0	11.5	13.0	13.3	13.1	12.8	12.9	11.6	10.4	7.7	5.8		
		MCWBR	4.6	5.8	7.4	7.0	6.6	6.1	5.4	5.4	6.0	6.3	5.4	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.345	0.384	0.436	0.429	0.427	0.448	0.437	0.402	0.374	0.348	0.314		
	taud		2.303	2.279	2.219	2.129	2.196	2.238	2.191	2.228	2.294	2.336	2.353	2.350		
	Ebn at Noon		696	774	806	803	828	833	806	794	780	730	655	646		
	Edh at Noon		68	90	115	140	137	133	137	126	106	85	66	57		
(44) All-Sky Solar Radiation	RadAvg		0.81	1.53	2.63	4.09	5.00	5.27	5.01	4.50	3.20	1.88	0.92	0.62		
	RadStd		0.15	0.26	0.39	0.53	0.59	0.52	0.60	0.39	0.46	0.27	0.13	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.51	N/A	N/A	+0.80	N/A	+0.38	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+0.48	N/A	N/A	+0.73	N/A	N/A	N/A	N/A	+59	+22

Nomenclature: See separate page